

ICP Pipeline Milestone 1 Report

Pipeline Overview

- Pipeline: ICP ID Card OCR
- Objective: Classify which country an ID card belongs to
- Architecture summary: The pipeline has 3 stages: image enhancement, channel selection, and OCR. Each pipeline step described below has a contract for input and output types such that each step in the pipeline can be combined to run with any other configuration of the pipeline. Every combination of the pipeline is then run and accuracy assertions are made.

Architecture Detail

Dataset

- `lansinote/ocr_id_card` from HuggingFace
 - ID cards, written in Chinese

Stage 1: Image Enhancement

- Enhancement of base image, e.g. brightness or contrast. This is done with cv2 library.

Stage 2: Channel Selection

- Extraction of a specific color from the image, e.g. red or green, blue. This is done with cv2 library.

Stage 3: OCR

- Optical character recognition that converts image text to true text. This is done with EasyOCR and Tesseract.

Assertions / accuracy metrics

- Accuracy is measured with four assertions that are checked on the OCR output of each card. The assertions do not use the answer key, so they can also be applied to new cards.
 1. Date valid: the OCR output contains a year between 1900 and 2030 followed by a month and a day, and the three numbers form a real calendar date. This is the example assertion provided with the pipeline.
 2. ID format: the OCR output contains an 18-digit ID number.

3. ID matches date: digits 7 to 14 of the ID number equal the date from assertion 1. In this dataset these digits are the birth date on all 20,000 cards, so a mismatch means one of the two was misread.
 4. Sex valid: the OCR output contains exactly one of 男 (male) or 女 (female).
- The accuracy of a configuration is the number of assertions that pass, summed over its cards. With 20 cards per configuration the maximum is 80. The accuracy (%) column in the results table gives the same count as a percentage of the maximum.
 - A second metric compares the OCR output to the dataset's answer key. A field counts as correct if its true value appears in the OCR text. For year, month, and day, the value must appear as a separate number outside the ID number, and each number in the OCR text can only be matched once. This rule is needed because digits 7 to 14 of every ID number are the birth date, so without it the year, month, and day would also be found inside the ID number. A stray number elsewhere in the OCR text can still match one of these values by chance. The results table reports the percentage of the 8 fields that were read correctly. This metric is included because an output can pass the assertions and still be wrong, for example a valid date that is not the date on the card. Ethnicity is 汉 on every card in the dataset, so for that field the metric only shows whether the OCR read that character.

Machine Configuration on Chameleon

- Site: KVM@TACC
 - Provisions a VM quickly and is CPU-only, which suffices since neither EasyOCR nor Tesseract requires a GPU
- Flavor: M1.Medium
 - Enough to load EasyOCR's and other required models
- Image: CC-Ubuntu24.04
 - Chameleon supported and useful for cleanly installing docker

All Configurations

- Number of datasets: 1 (lansinuoote/ocr_id_card, 20,000 ID card images)
- Number of enhancement options: 12 (none, +.25 contrast, +.50 contrast, +.75 contrast, -.50 brightness, +50 brightness, +100 brightness, simple laplacian, precise laplacian, +50% saturation, sharpen, deskew)
- Number channel options: 8 (none, blue, green, red, red-green, green-blue, red-blue, grayscale)
- Number of OCR options: 2
- Total distinct pipeline configurations = $12 \times 8 \times 2 = 192$

- Cards per configuration: 20, chosen at random from the 20,000 cards with a fixed seed (42) so that every run uses the same cards, for 3,840 pipeline runs in total
- Running all 192 configurations on all 20,000 cards would take about 1,620 hours at the measured speeds of 2.41 seconds per card for EasyOCR and 0.62 seconds per card for Tesseract.

Results Table

The results come from building the team Dockerfile into a container and running the notebook inside it on a Chameleon m1.medium instance with 2 vCPUs and 4 GB of memory.

The notebook (milestone_1.ipynb) and the Dockerfile are in the team repository:

https://github.com/ashleyogorman/ICP_Milestone_1

The table shows the 10 configurations with the highest accuracy and the Tesseract baseline. A baseline uses no enhancement and no channel selection. The EasyOCR baseline is in the top 10 at rank 8, and the Tesseract baseline is at rank 91. The rank column gives the position of each configuration out of all 192, sorted by accuracy and then by fields correct.

The full table for all 192 configurations is in results/results_table.csv:

https://github.com/ashleyogorman/ICP_Milestone_1/blob/main/results/results_table.csv

The results for each card are in results/card_results.csv:

https://github.com/ashleyogorman/ICP_Milestone_1/blob/main/results/card_results.csv

A second run of the same container on a new Chameleon instance gave the same assertion and field results for all 192 configurations. Its mean latencies differed by at most 0.12 seconds per configuration.

The SLAs for the pipelines have not been released yet, so the SLA met column is marked TBD.

Table 1. Top 10 configurations and the Tesseract baseline, 20 cards per configuration

Rank	Enhancement	Channel	OCR	Assertions passed (of 80)	Accuracy (%)	Fields correct (%)	Mean latency (s)	SLA met
1	deskew	green-blue	EasyOCR	58	72.5	64.4	2.39	TBD
2	deskew	none	EasyOCR	56	70.0	62.5	2.53	TBD
3	sharpen	red	Tesseract	56	70.0	61.9	0.65	TBD
4	deskew	blue	EasyOCR	55	68.8	63.1	2.34	TBD
5	-50 brightness	blue	EasyOCR	53	66.2	66.2	2.31	TBD
6	deskew	red	Tesseract	53	66.2	66.2	0.65	TBD
7	deskew	blue	Tesseract	52	65.0	66.9	0.55	TBD
8	none	none	EasyOCR	52	65.0	65.6	2.55	TBD
9	+50% saturation	blue	EasyOCR	51	63.8	63.1	2.33	TBD
10	deskew	red-green	Tesseract	51	63.8	62.5	0.62	TBD
91	none	none	Tesseract	39	48.8	54.4	0.68	TBD

The best configuration was deskew, green-blue, and EasyOCR, which passed 58 of 80 assertions (72.5%). The EasyOCR baseline passed 52 of 80 and the Tesseract baseline passed 39 of 80. Deskew had the highest average of the 12 enhancement options, at 47.8 of 80 assertions over its 16 configurations, compared with 40.7 for no enhancement. Sharpen, +.25 contrast, and +50 brightness also averaged more than no enhancement, and the other seven options averaged fewer. The largest drops were for +100 brightness at 29.0, +.75 contrast at 25.3, simple Laplacian at 7.6, and precise Laplacian at 0.4 of 80. The averages for the 8 channel options ranged from 31.6 to 35.0 of 80, so the choice of channel made little difference. No channel selection had the highest average. EasyOCR and Tesseract read about the same share of fields correctly over all configurations, 44.1% and 43.8%. Tesseract was about four times faster than EasyOCR, at 0.62 seconds per card compared with 2.41 seconds.

Team Organization

The team organized itself such that each team member handled 3-5 image enhancement and channel selection options and 1 OCR pipeline option. We communicated via text message and had bi-weekly meetings via Teams. We did not encounter any teamwork problems, though in the next milestone we hope to target an earlier deliverable date.